

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – VI

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Part B - Program

- a) **Create a simple website for a College Information Portal and convert it into a Progressive Web Application.**

Algorithm :

STEP 1 :- START

STEP 2 :- Create manifest.json and service-worker.js, and add the required app icons.

STEP 3 :- Link the manifest in Layout.cshtml and register the Service Worker using JavaScript.

STEP 4 :- STOP

Code:

manifest.json :-

```
{
  "name": "HTML TO PWA",
  "short_name": "HTP",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "128/128",
      "type": "image/png",
      "purpose": "any"
    }
  ],
}
```

```
{
  "src": "icons/icon2.png",
  "sizes": "512/512",
  "type": "image/png",
  "purpose": "any"
}
]
```

service-worker.js :-

```
self.addEventListener("Install", e => {
  console.log("Service Worker Installed");
});
```

```
self.addEventListener("fetch", e => {
  e.responseWith(fetch(e.request));
});
```

Layout.cshtml :-

Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

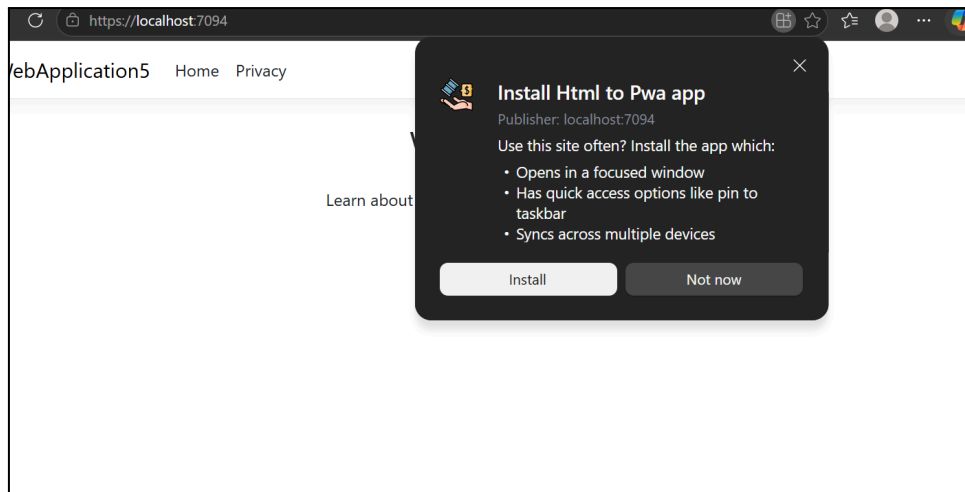
Before the closing </body> tag:

```
<script>
```

```
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/service-worker.js');
}
```

```
</script>
```

Output:



b) Create a PWA that displays a custom offline page when the user is not connected to the internet.

Algorithm :-

STEP 1 :- START

STEP 2 :- Create offline page with the message

STEP 3 :- Configure the service worker so it loads the offline page

STEP 4 :- STOP

Code:

Offline.html :-

```
<!DOCTYPE html>
<html>
<head>
  <title>Offline</title>
</head>
<body>
  <h1>You are Offline</h1>
  <p>Please check your internet connection.</p>
</body>
</html>
```

Output:

You are Offline

Please check your internet connection.